

Julia Fang

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Education

Churchill College, University of Cambridge | MEng Engineering

Oct 2024 - Jun 2028

Second Year Part IB: Class I

Third Year Part IIA modules include:

- semiconductor engineering, dynamics, vibrations, machine design, signals and systems, systems and control, statistical signal processing, inference, medical imaging and 3D computer graphics, mathematical methods.

A-Level Mathematics (A*), A-Level Further Mathematics (A*), A-Level Physics (A*); achieved the **highest mark in the world** for Pearson Edexcel IAL Mathematics.

Experience & Projects

Summer internship project supervised by Alice Cicirello and Ramit Debnath

Jun - Sep 2026

- Interdisciplinary project merging AI, arts, humanities, and engineering.
- An environmental-sensing LED installation for Churchill College, integrating machine learning to transform a physical space into a living hub for exploring human-computer-environment interaction (HCE).

Integrated Design Project

2026

- Design, build and test a mobile robot vehicle in a team of seven; worked in the software subteam on sensor integration, decision logic for vehicle movement and pathfinding algorithm.

Riv-Link Manufacturing Intern

Jul 2025

- Authored documentation on new products including control plan (CP), standard of procedure (SOP) for manufacturing process and quality control.

Cambridge University Space Flight Society

2024 - 2025

- Working in the propulsion team to design a 3kN max thrust throttleable rocket engine for Landers Challenge.

Cambridge University Locomotive Engineers' Society

2024-2025

- Structural and dynamics design for a miniature train as part of the Railway Challenge.

Industrial experience at ZEnergy Battery Technologies

Dec 2024

- Electrical vehicle battery manufacturing process and business operation: industrial chain, trends and prospects.

Research project: Efficiency and sustainability of transportation modes considering modern day statistics (2023)

- Built an interactive questionnaire in code that recommends an optimal transport mode from live statistics.

Research project: Vibration control of architectural structure in an earthquake (2022)

- Self-directed research inspired by grandparents' experience with minor earthquakes, awarded High Distinction for the Young Scientist Awards.

Awards and Competitions

- International Mathematics Modelling Challenge (IMMC): national finalist for consecutive years (2021-2024), honourable mention
- Australian Intermediate Mathematics Olympiad, Physics and Chemistry Olympiads: High Distinction

Additional Skills

IT: Microsoft Office, Python, CAD Solidworks and Fusion, C++

Languages: English, Chinese, French [Intermediate]

Interests: Tennis (9 years), Dance | Choir (8 years) | Duke of Edinburgh: Silver

Referees

Prof. Seb Savory, Director of Studies (Part I), Churchill College, University of Cambridge sjs1001@cam.ac.uk

Dr Andrew Taylor, Tutor, Churchill College, University of Cambridge awt24@cam.ac.uk